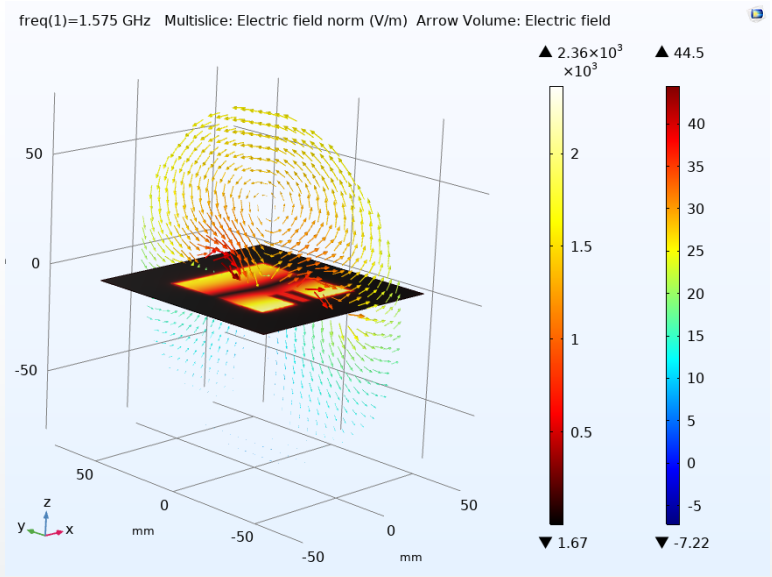


Fizica III - Electromagnetism

Suport de curs, A&C, anul II, F3, 2022-2023

Prof.dr.ing. Gabriela Ciuprina gabriela@lmn.pub.ro – C, P.

As.dr.ing. Mihai Popescu mihai_p@lmn.pub.ro – P.

S.l..dr.ing. Sorin Lup sorin@lmn.pub.ro – P.

0. Introducere

- 0.0. Chestiuni organizatorice
- 0.1. Obiectivul cursului
- 0.2. Electromagnetism: scurt istoric si aplicatii
- 0.3. Structura cursului

0.0. Chestiuni organizatorice

1 C (marti, saptamana para, 14.00-16.00, Precis 106)

1 P(Ldemo,S) (3 grupe, Precis 106:

a) marti, impar, 12.00-14.00;

b) marti, impar, 14.00-16.00;

c) marti, par, 12.00-14.00)

- C, P: Prof.dr.ing. Gabriela Ciuprina gabriela@lmn.pub.ro sau gabriela.ciuprina@upb.ro, chat TEAMS (ID: 24546), birou EAD 04
- P: As.dr.ing. Mihai Popescu mihai_p@lmn.pub.ro sau mihai.popescu@upb.ro, chat TEAMS (ID: 23822), birou EAD 05
- P: S.l..dr.ing. Sorin Lup sorin@lmn.pub.ro sau sorin.lup@upb.ro, chat TEAMS (ID: 77056), birou EAD 07

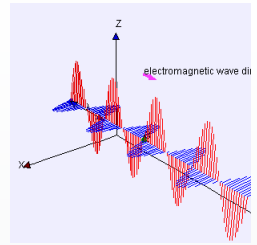
Notare:

- in timpul semestrului: 20% curs, 80 % aplicatii.
- verificare pe parcurs, nota se incheie in saptamana 14.

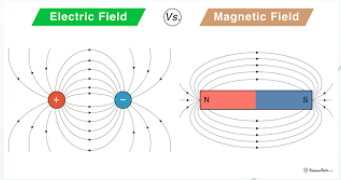
Pagina moodle (materiale didactice, referinte, anunturi, quiz-uri):

- https://curs.upb.ro/2022/course/view.php?id=5288

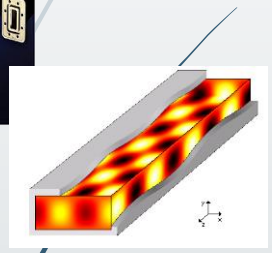
0.1. Obiectivul cursului (din fisa disciplinei)



https://en.wikipedia.org/wiki/Electromagnetic_radiation#/media/File:Electromagneticwave3D.gif





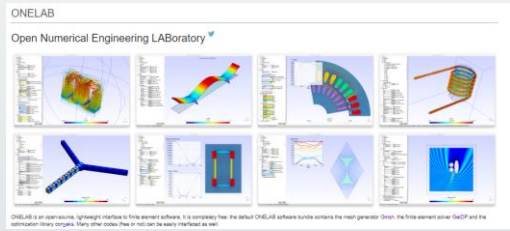


https://en.wikipedia.org/wiki/Waveguide#/media/File:Waveguide_x_EM_rect_TE31.gi f

- Aprofundarea principalelor concepte pe care se bazeaza teoria câmpului electromagnetic și intelegerea unor aplicatii in care acestea intervin ca de exemplu:

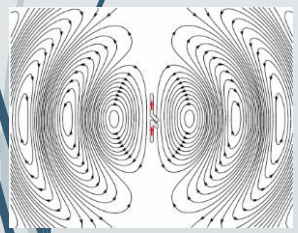
1) proiectarea circuitelor integrate (VLSI), analiza fenomenelor parazite;
2) propagarea undelor in mod ghidat prin interconexiuni, cabluri Ethernet, ghiduri de unda, fibre optice, sau
3) propagarea undelor in mod neghidat: unde radio, WiFi, bluetooth.



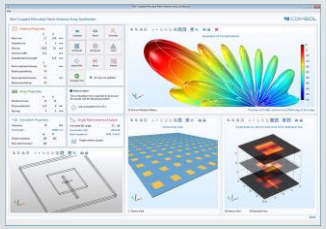


https://onelab.info/

- Evidentierea importanței modelării în inginerie.
- Evidentierea legaturilor cu alte discipline facute anterior (Metode Numerice, Algoritmi si structuri de date), in curs de desfasurare (Electronica), sau viitoare, Calcule stiintifice de inalta performanta).

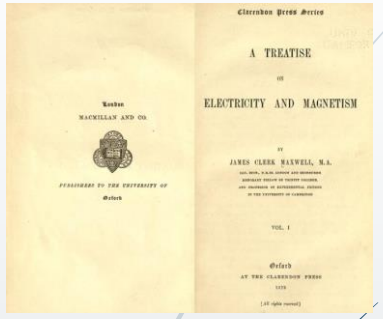


https://commons.wikimedia.org/wiki/File:Dipole_xmting_antenna_animation_4_408x318x150ms.gif

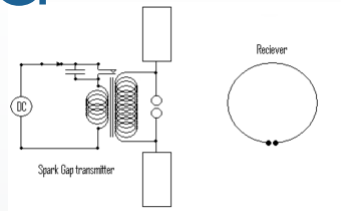


https://www.comsol.com/release/5.2/rf-module

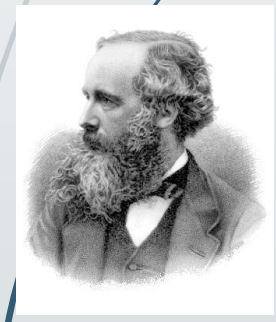
0.1. Obiectivul cursului (de ce EM la A&C ?)







- Campul EM este o realitate fizica cu care suntem in contact permanent, desi nu avem simturi pentru a o percepe.
- Cea mai pura forma de manifestare a campului EM, undele EM, au fost descoperite in 1893 (acum 130 de ani).
- Nu ne putem imagina viata astazi fara inventiile bazate pe fenomenele EM.
- Teoria campului EM face parte din cultura umanitatii.



James Clerk Maxwell (1831-1879)

https://en.wikipedia.org/wiki/James_Clerk_Maxwell

1864

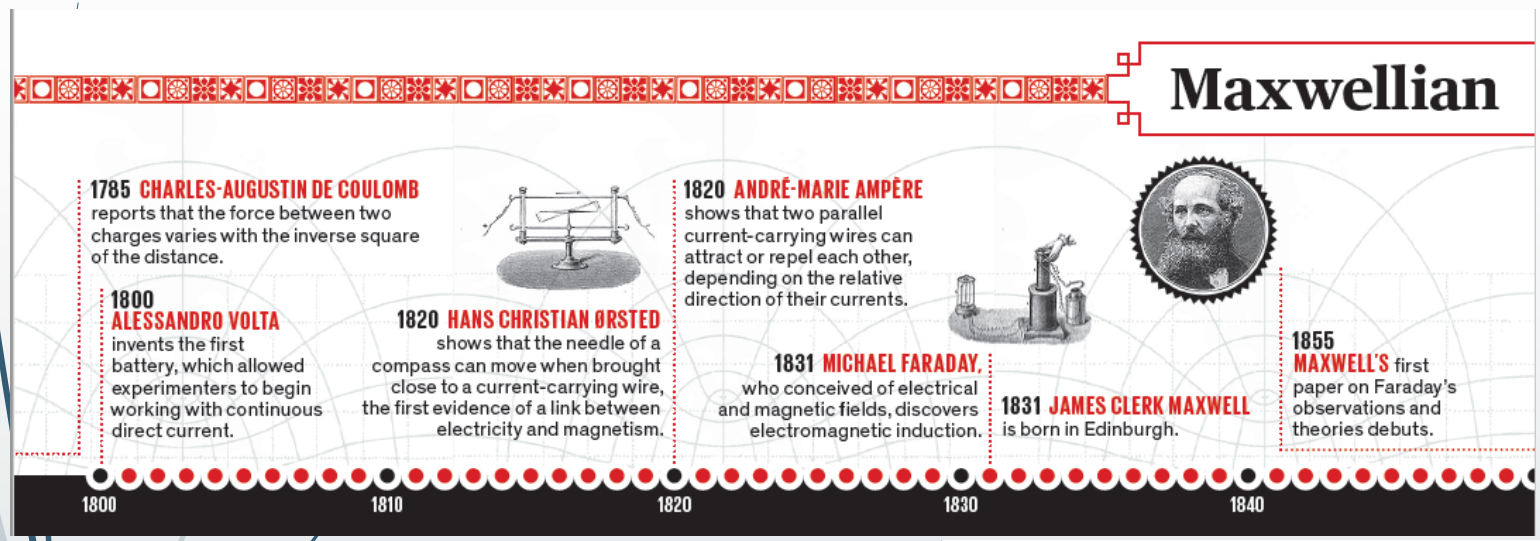


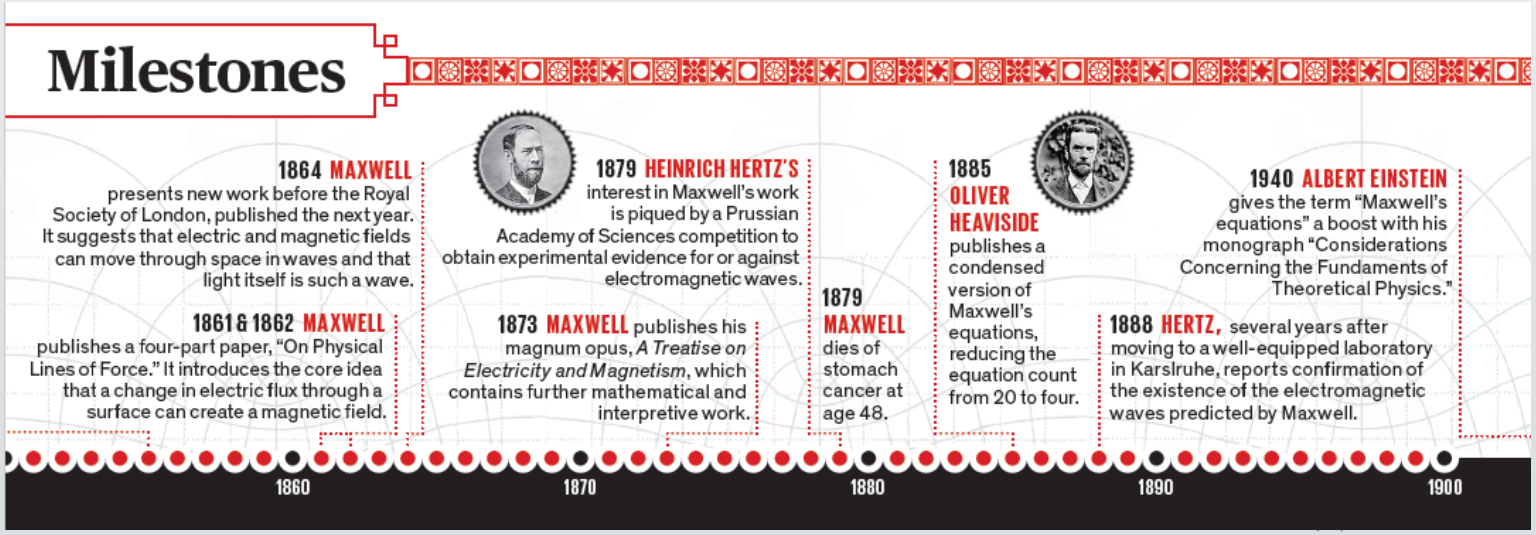
Heinrich Rudolf Hertz (1857-1894)

https://en.wikipedia.org/wiki/Heinrich_Hertz

1887-1889

0.2. Electromagnetism: scurt istoric

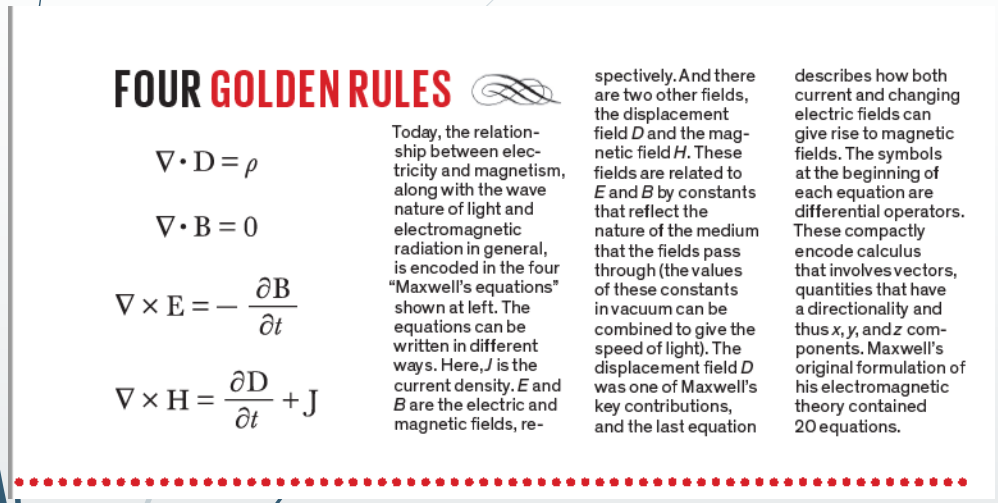




J. C. Rautio, "The Long Road to Maxwell's Equations," in *IEEE Spectrum*, vol. 51, no. 12, pp. 36-56, December 2014, doi: 10.1109/MSPEC.2014.6964925.

(disponibil pe moodle)

0.2. Electromagnetism: scurt istoric



FOUR GOLDEN RULES

$$\nabla \cdot \mathrm{D} = \rho$$

$$\nabla \cdot \mathrm{B} = 0$$

$$\nabla \times \mathrm{E} = -\frac{\partial \mathrm{B}}{\partial t}$$

$$\nabla \times \mathrm{H} = \frac{\partial \mathrm{D}}{\partial t} + \mathrm{J}$$

Today, the relationship between electricity and magnetism, along with the wave nature of light and electromagnetic radiation in general, is encoded in the four "Maxwell's equations" shown at left. The equations can be written in different ways. Here, *J* is the current density. *E* and *B* are the electric and magnetic fields, respectively. And there are two other fields, the displacement field *D* and the magnetic field *H*. These fields are related to *E* and *B* by constants that reflect the nature of the medium that the fields pass through (the values of these constants in vacuum can be combined to give the speed of light). The displacement field *D* was one of Maxwell's key contributions, and the last equation describes how both current and changing electric fields can give rise to magnetic fields. The symbols at the beginning of each equation are differential operators. These compactly encode calculus that involves vectors, quantities that have a directionality and thus *x*, *y*, and *z* components. Maxwell's original formulation of his electromagnetic theory contained 20 equations.

Name	Equation	
	Integral form	Differential form
Faraday's law of induction	$\oint_c \vec{E} \cdot d\vec{l} = -\iint_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$
Ampère-Maxwell law	$\oint_c \vec{H} \cdot d\vec{l} = \iint_S \vec{J} \cdot d\vec{S} + \iint_S \frac{\partial \vec{D}}{\partial t} \cdot d\vec{S}$	$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$
Gauss' electric law	$\oiint_S \vec{D} \cdot d\vec{S} = \iiint_V \rho \, dV$	$\nabla \cdot \vec{D} = \rho$
Gauss' magnetic law	$\oiint_S \vec{B} \cdot d\vec{S} = 0$	$\nabla \cdot \vec{B} = 0$

J. C. Rautio, "The Long Road to Maxwell's Equations," in *IEEE Spectrum*, vol. 51, no. 12, pp. 36-56, December 2014, doi: 10.1109/MSPEC.2014.6964925.

(disponibil pe moodle)

Cateva din obiective specifice ale acestei discipline

- Dezvoltarea **limbajului** specific teoriei campului EM
- Cunoasterea **marimilor fizice** (semnificatie, unitati si metode de masura).
- Intelegerea **semnificatiei fizice** ale legilor EM (**fenomene**).
- Intelegerea principalelor **teoreme** ale EM.
- Intelegerea principalelor **aplicatii practice** care folosesc camp EM.
- Deschiderea catre **aspecte algoritmice.**

Electromagnetism: aplicatii

Hertz nu a realizat importanta practica a experimentelor sale cu unde EM.
Intrebat de studenti la ce folosesc undele EM, a raspuns https://spark.iop.org/hertzs-useless-discovery

"*It's of no use whatsoever*[...] *this is just an experiment that proves Maestro Maxwell was right—we just have these mysterious electromagnetic waves that we cannot see with the naked eye. But they are there.*"

"no use" ?

0.2. Electromagnetism: aplicatii

Pentru o lista a principalelor inventii ce folosesc fenomene EM puteti consulta de exemplu

D.Ioan, Bazele electrotehnicii, 2012. http://www.lmn.pub.ro/~daniel/BazeELTH-0-Intro.pdf

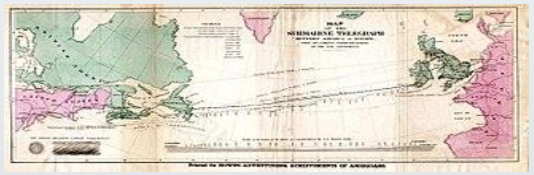
Cateva exemple:

1837 **Samuel Morse** Breveteaza telegraful electric, system de transmitere a mesajelor text



https://en.wikipedia.org/wiki/Electrical_telegraph#Morse_System

1858 **Cyrus West Field** Primul cablu transatlantic



https://en.wikipedia.org/wiki/Transatlantic_telegraph_cable

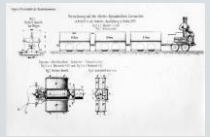
1876 **Alexander Graham Bell** Breveteaza telefonul



https://en.wikipedia.org/wiki/Alexander_Graham_Bell

Siemens

1879 Construieste primul tren electric de pasageri



https://new.siemens.com/global/en/company/about/history/stories/on-track.html

1880 Construieste primul ascensor electric



https://new.siemens.com/global/en/company/about/history/stories/electric-elevator.html

1881 Instaleaza primul tramvai electric



https://new.siemens.com/global/en/company/about/history/stories/first-electric-streetcar.html

0.2. Electromagnetism: aplicatii

1882 Edison Instaleaza prima retea energetica de c.c.

1885 Nicola Tesla Prima retea energetica de curent alternativ

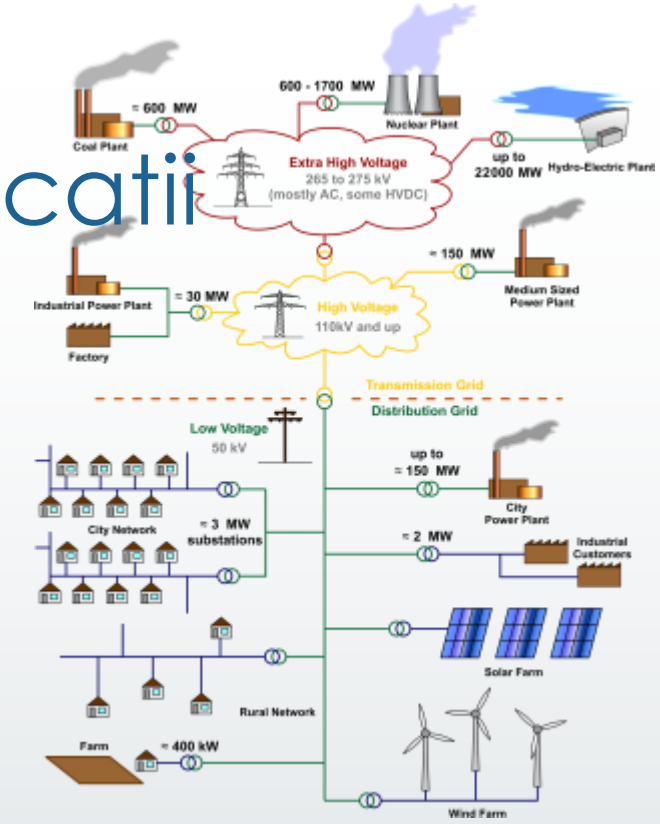
1888 Westinghouse Prima retea trifazata

https://en.wikipedia.org/wiki/War_of_the_currents#Edison's_DC_company

1888 Nicola Tesla Breveteaza motorul asincron (cu inductie)

http://ffden-2.phys.uaf.edu/211_fall2013.web.dir/Ryker_Dial/The_Current_Wars/AC_Motor.html



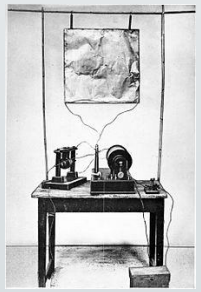


https://en.wikipedia.org/wiki/Electrical_grid

1897 Nicola Tesla Breveteaza radioul

https://en.wikipedia.org/wiki/Invention_of_radio

1901 Guglielmo Marconi Prima transmisie radio transatlantica



https://en.wikipedia.org/wiki/Guglielmo_Marconi

1931 Edwin Howard Armstrong Dezvolta tehnologia pentru televiziunea electronica

https://en.wikipedia.org/wiki/Edwin_Howard_Armstrong

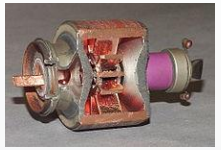
0.2. Electromagnetism: aplicatii



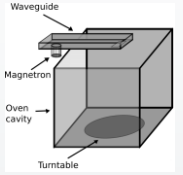
https://ro.wikipedia.org/wiki/RADAR

1940 **Randall si Boot**

Inventeaza tubul electronic numit magnetron, care permite generarea microundelor, ceea ce a condus la dezvoltarea Radarului si cuptoarelor cu microunde



https://en.wikipedia.org/wiki/Cavity_magnetron





https://en.wikipedia.org/wiki/Microwave_oven

1941 **Konrad Zuse**

Inventeaza Z3 –primul calculator programabil deplin functional, cu 2000 relee si frecventa

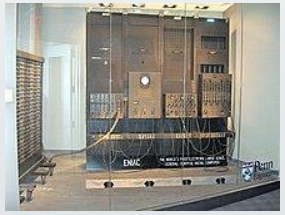
ceasului de 10 Hz (cuvantul de 22 biti).



https://ro.wikipedia.org/wiki/Z3_(computer)

1946 **John Presper Eckert si John Mauchly**

Pun in functiune ENIAC(Electronic Numerical Integrator and Computer), primul calculator electronic digital, ce contine17,468 tuburi, 1500 relee cu frecventa de lucru de 5000Hz si lungimea cuvantului de 10 biti.



https://en.wikipedia.org/wiki/ENIAC

1947 **JohnBardeen si Walter Brattain**

Inventeaza tranzistorul, care abia in 1957 se produce in serie si se realizeaza primul radio-tranzistor.



https://ro.wikipedia.org/wiki/Tranzistor

1958 **Jack Kilby si Robert Noyce**

Inventeaza circuitele integrate



https://en.wikipedia.org/wiki/Integrated_circuit

0.2. Electromagnetism: aplicatii

1968 **Marcian Hoff**

Inventeaza microprocesorul, dar abia in 1974 se produce Intel 8080, microprocesorul de 8 biti, care a stat la baza primului calculator personal (cu frecventa ceasului de 2MHz), ALTAIR,

produs comercial in 1975.

https://engineering.stanford.edu/news/ted-hoff-birth-microprocessor-and-beyond

1969 **Robert H. Dennard**

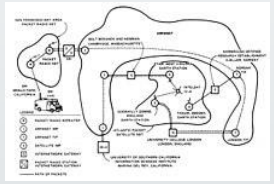
Inventeaza memoria RAM dinamica (DRAM) cu un transistor, permitand o crestere importanta de viteza si capcitate de memorare. Este o componenta importanta a memoriei calculatoarelor si in ziua de azi, cand se produc cipuri de 1Gbit pentru frecvente de peste 1GHz.

https://en.wikipedia.org/wiki/Dynamic_random-access_memory

1970 **Vinton Cerf**
Robert Kahn

Definesc TCP/IP (Transmission Control Protocol / Internet Protocol), care devine in 1983 unicul protocol oficial al Internetului,

conducand la explozia sa.



https://en.wikipedia.org/wiki/Internet_protocol_suite

1973 **Martin Cooper**

Telefonul mobil, prezentat in forma comerciala

de Motorola abia in 1983.



https://en.wikipedia.org/wiki/Mobile_phone

1979 **Allan McLeod Cormacksi Godfrey Hounsfield**

Inventia Computer-tomografulului cu raze X, pentru care au luat Premiul Nobel in Medicina

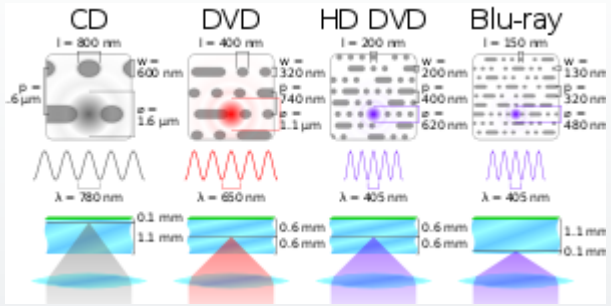


https://en.wikipedia.org/wiki/CT_scan

0.2. Electromagnetism: aplicatii

1979 **Sony si Philips**

Proiectarea discului compact digital CD optic. Primele aparate de reproducere audio sunt disponibile comercial in 1982. Marcheaza inceputul revolutiei digitale.



https://en.wikipedia.org/wiki/Compact_disc

1981 **Epson, Tandy si HP**

Produc primele calcualtoare portabile, de tip laptop, dar rezolutie VGA este disponibila abia in 1988, dupa dezvoltarea tehnologiei LCD (Liquid-crystal display).

1983 **International Telecommunication Union' (ITU-R)**

A initiat un standard international unic pentru telviziunea digitala de inalta definitie HDTV. Standardul este folosit atat pentru transmisia prin satelit, cablu, receptia si inregistrarea semnalelor video. Prima emisie a avut loc in Belgia de canalul HD1, care a transmis in 2004 concetul de anul nou de la Viena.

1991 **Radiolinja, Telecom Finlanda**

Telefonul cellular automat de generatia 2, conform standarduluiGSM (Global System for communications).

Generation	Technology	Feature	Encoding	Year of First Use	Roaming	Handset interoperability	Common Interference	Signal quality/coverage area	Frequency utilization/Call density	Handoff	Voice and Data at th time
1G	FDMA	NMT	Analog	1981	Nordics and several other European countries	None	None	Good coverage due to low frequencies	Very low density	Hard	No
2G	TDMA and FDMA	GSM	Digital	1991	Worldwide, all countries except Japan and South Korea	SIM card	Some electronics, e.g. amplifiers	Good coverage indoors on 850/900 MHz. Repeaters possible. 35 km hard limit.	Very low density	Hard	Yes GPRS Clas
2G	CDMA	IS-95 (CDMA one)	Digital	1995	Limited	None	None	Unlimited cell size, low transmitter power permits large cells	Very low density	Soft	No
3G	CDMA	IS-2000 (CDMA 2000)	Digital	2000 / 2002	Limited	RUIM (rarely used)	None	Unlimited cell size, low transmitter power permits large cells	Very low density	Soft	No EVDO / Yes SV
3G	W-CDMA	UMTS (3GSM)	Digital	2001	Worldwide	SIM card	None	Smaller cells and lower indoors coverage on 2100 MHz; equivalent coverage indoors and superior range to GSM on 850/900 MHz.	Very low density	Soft	Yes[3]
4G	OFDMA	LTE	Digital	2009	Worldwide	SIM card	None	Smaller cells and lower coverage on the S band.	Very low density	Hard	No (data only Voice possible throug or fallback to 2G
5G	OFDMA	NR	Digital	2018	Limited	SIM card	None	Dense cells on millimeter waves.	Very low density	Hard	No (data only Voice possible throug

https://en.wikipedia.org/wiki/Comparison_of_mobile_phone_standards

0.2. Electromagnetism: aplicatii

1994 **US – Department of Defence**

Folosirea civila a sistemului de pozitionare globala GPS.

1995 **Philips,Sony,Toshibas iPanasonic**

Inventeaza si dezvolta DVD-ul-disc optic de mare capacitate pentru stocarea datelor digitale, ce permite inregistrarea video, chiar daca acesta are aceleasi dimensiuni cu discul CD audio.



Astazi, aplicatiile din EM (camp, circuite electronice/digitale) sunt in conexiune cu aplicatii din alte domenii fizice (mecanic, fluide, termodinamic, etc) => sisteme integrate

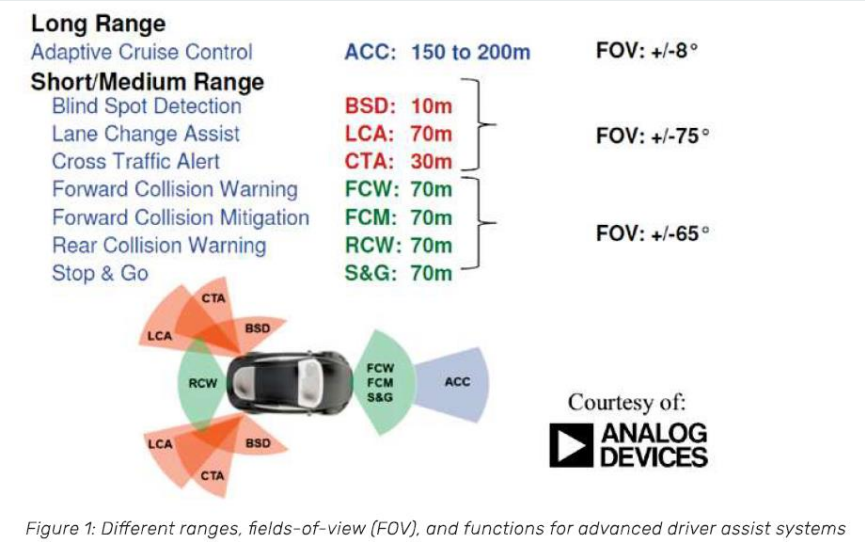
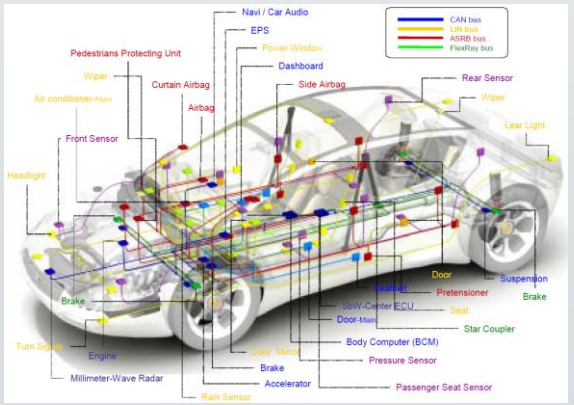
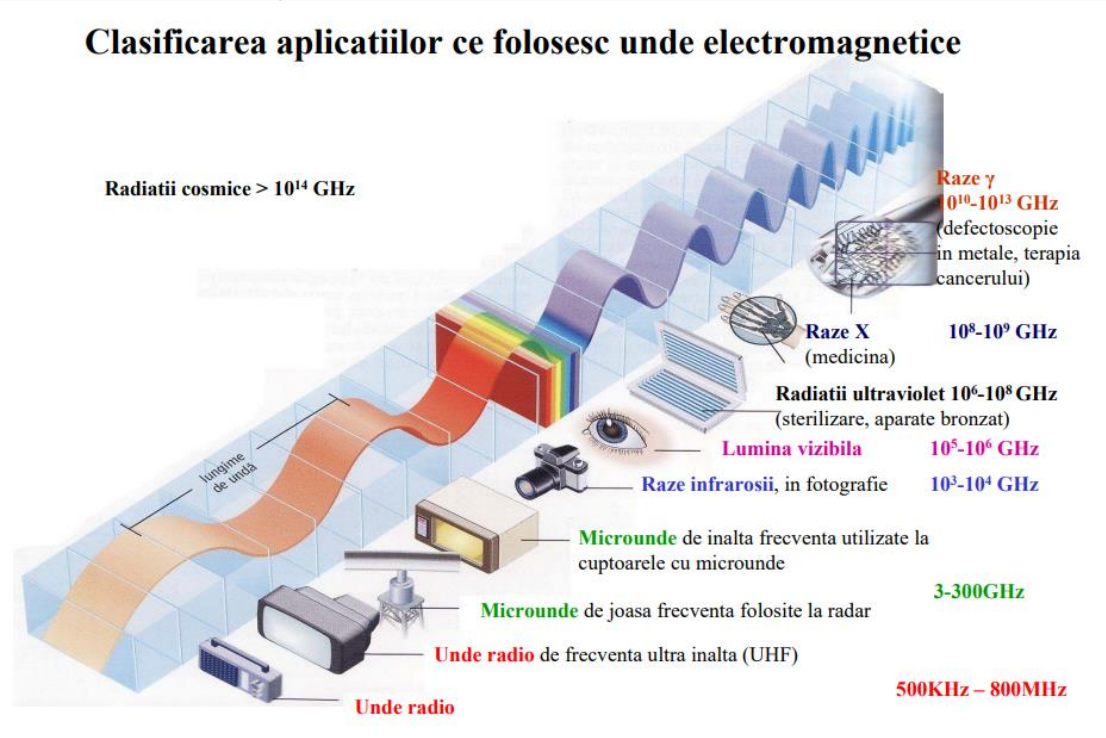
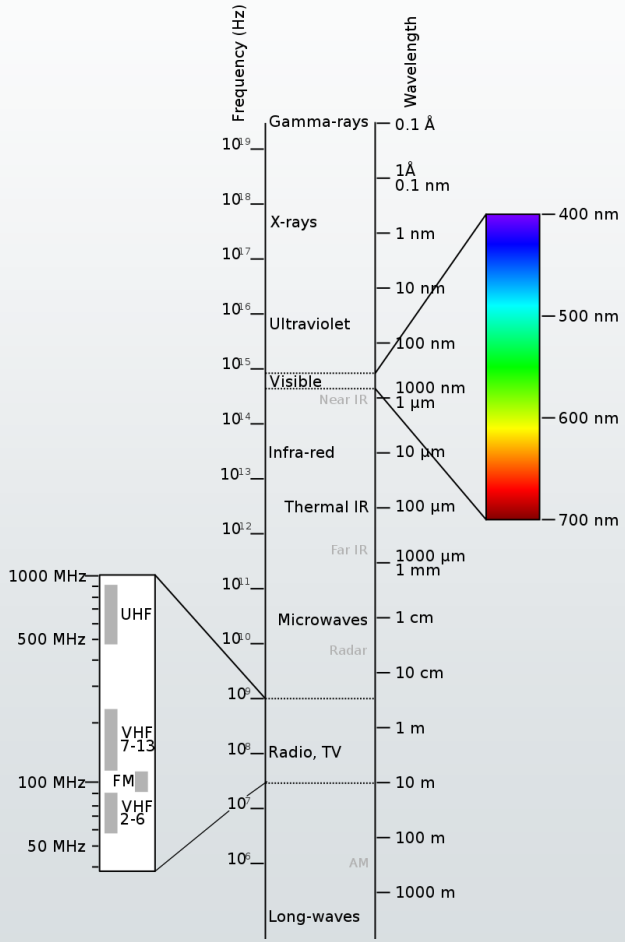


Figure 1: Different ranges, fields-of-view (FOV), and functions for advanced driver assist systems



0.2. Electromagnetism: aplicatii





https://en.wikipedia.org/wiki/Electromagnetic_spectrum

0.3. Structura cursului – pe scurt

Cateva din obiectivele specifice:

- Dezvoltarea **limbajului** specific teoriei campului EM
- Cunoasterea **marimilor fizice** (semnificatie, unitati si metode de masura).
- Intelegerea **semnificatiei fizice** ale legilor EM (**fenomene**).
- Intelegerea principalelor **teoreme** ale EM.
- Intelegerea unor **aplicatii practice** care folosesc camp EM.
- Deschiderea catre **aspecte algoritmice.**

- Curs 1. Obiectivul cursului.
  Marimile campului EM.
- Curs 2. Legile EM.
- Curs 3. Teoremele EM.
- Curs 4. Circuite electrice cu parametri distribuiti.
- Curs 5. Unda EM plana. Propagare.
- Curs 6. Emisia si receptia undelor. Antene.
- Curs 7. Analiza numerica a campului EM.

Cursurile sunt intercalate de aplicatii (P dar poate fi si de tip L-demonstrative sau S). Fiecare sedinta de aplicatie va tinti ilustrarea conceptelor prezentate la cursurile anterioare, tintind in final transmisia si receptia informatiei prin cablu sau prin aer. Tematica fiecarei saptamani (include referinte), va fi actualizata pe moodle.